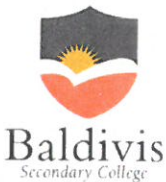


Name:			Date:	
Teacher:	SOLUTIONS			
	Year 10 Mathematics Mini test 6 Topic – Pythagoras		SCORE: / 21	
	<u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	35 minutes			
Weighting:	5 %			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, ruler and a A4 page of notes.			

Multi-choice Questions;

(5 marks)

Question1.

Q. What is the longest side of a right triangle called?

answer choices

- ☐ Leg A
 ☐ Leg B
 ☒ Hypotenuse
 ☐ Pythagoras' Big Buddy

Question2.

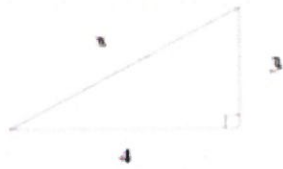
Q. What is the formula used to solve for the hypotenuse?



answer choices

- ☒ $a^2 + b^2 = c^2$
☐ $a + b = c$
☐ $a^2 - b^2 = c^2$
☐ $c - a = c$

Question 3.



Q. What is the length of x ?

answer choices

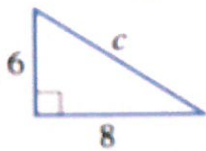
☒ 5

☐ 8

☐ 7

☐ 25

Question 4



Q. Use Pythagoras' Theorem to find c in this triangle:

answer choices

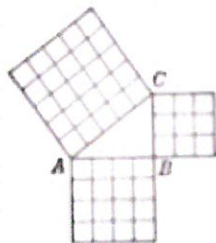
☐ 5.29

☐ 28

☒ 10

☐ 100

Question 5.



Q. What is the distance between A and C?

answer choices

☐ 25

☐ 4

☐ 3

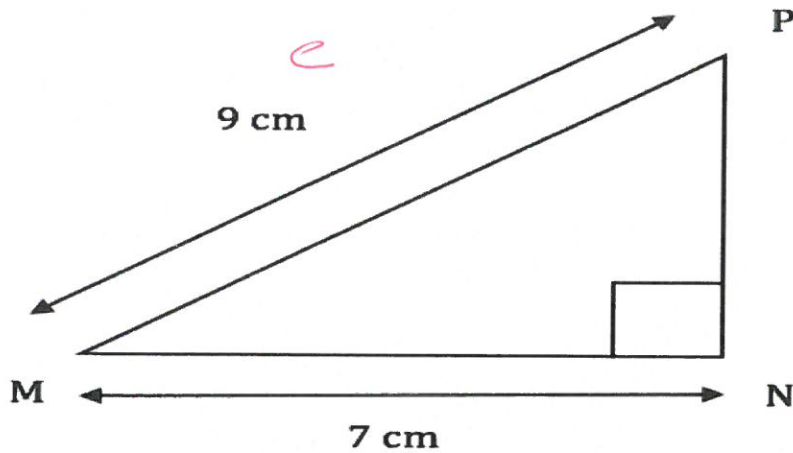
☒ 5

Short answer questions. SHOW ALL WORKINGS.

Question 6.

(3marks)

What is the length of the missing side?

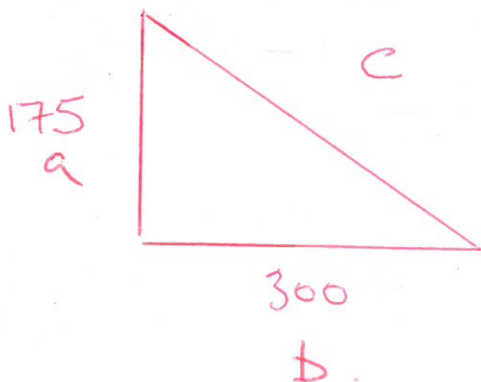


$$\begin{aligned}
 a &= \sqrt{c^2 - b^2} \\
 a &= \sqrt{9^2 - 7^2} \\
 a &= \sqrt{81 - 49} \\
 a &= \sqrt{35} \\
 a &= 5.9 \text{ cm}
 \end{aligned}$$

Question 7.

(4 marks)

David leaves the house to go to school. He walks 300m west and 175m north. How far away is he from his starting point? (the diagonal). Draw a diagram and show all workings.

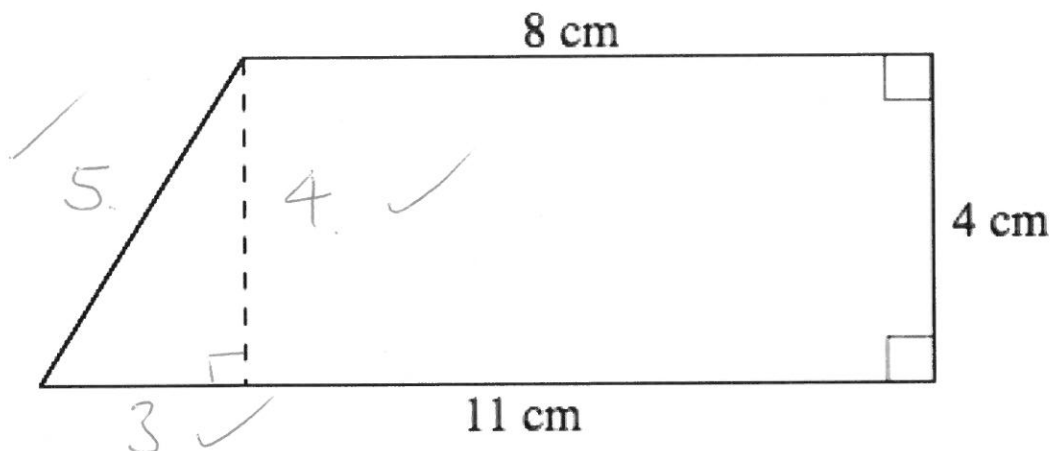


$$\begin{aligned}
 c^2 &= a^2 + b^2 \\
 c &= \sqrt{a^2 + b^2} \\
 c &= \sqrt{175^2 + 300^2} \\
 c &= \sqrt{120625} \\
 c &= 347.3 \text{ m}
 \end{aligned}$$

Question 8.

Calculate the perimeter of the following shape:

(5 marks)

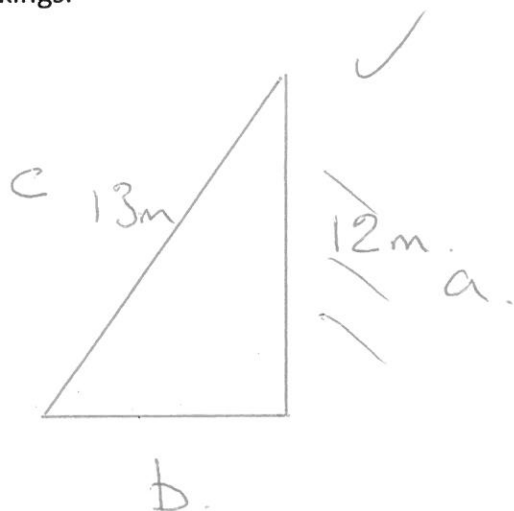


$$\text{Perimeter} = 28 \text{ cm} \quad \checkmark \checkmark$$

Question 9.

(4 marks)

If a fireman has a ladder that is 13 metres long. If he wants to reach a window that is 12 metres above the ground, how far from the wall should he put the bottom of his ladder? Draw a diagram and show all workings.



$$\begin{aligned} b &= \sqrt{c^2 - a^2} \quad \checkmark \\ b &= \sqrt{13^2 - 12^2} \quad \checkmark \\ b &= \sqrt{169 - 144} \\ b &= \sqrt{25} \\ b &= 5 \text{ m} \quad \checkmark \end{aligned}$$

The End